

# HILLOL DAS PIAS

AI-Native Full-Stack Engineer · 4.5+ Years Experience

+880 1738 206419 · hillol4958@gmail.com · Dhaka, Bangladesh

[Portfolio](#) · [GitHub](#) · [LinkedIn](#)

## PROFESSIONAL SUMMARY

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AI-native full-stack engineer with 4.5+ years building production web applications and a growing specialisation in LLM-powered product development. Hands-on with the OpenAI and Google Gemini APIs, LangChain, LangGraph, RAG pipelines (ChromaDB), tool-calling AI agents, multi-agent systems, and MCP integrations. Recently shipped a multi-agent customer support system featuring genuine human-in-the-loop control via LangGraph interrupt(), LLM-as-judge guardrails, and MCP-based tool execution that blocks any customer-facing action until human approval. Also built a deployed ReAct agent orchestrating 17 tools with LangSmith tracing, a test-gated CI/CD pipeline, and Docker-based deployment. Proficient across the modern JavaScript ecosystem (Next.js, Node.js, Express.js, React, TypeScript) and Python backends (FastAPI, async SQLAlchemy). Proven delivery across RESTful APIs, real-time systems (Socket.IO), headless CMS architectures and containerised services via Docker and CI/CD pipelines. Collaborative and self-driven, with experience across the full development lifecycle in remote-first environments.

## CORE COMPETENCIES

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**AI / LLM** LangChain · LangGraph · RAG (ChromaDB) · Multi-agent & ReAct agents · Human-in-the-loop (LangGraph interrupt()) · MCP · OpenAI & Gemini APIs · Tool use / function calling · Streaming · LangSmith · Prompt engineering · LLM-as-judge evaluation

**Full-Stack** Next.js · React · Node.js/Express · TypeScript · FastAPI · Socket.IO · REST APIs · Java (Spring Boot)

**Infra & Data** Docker · Docker Compose · GitHub Actions CI/CD · PostgreSQL · MySQL · Neon · async SQLAlchemy 2.0 · Alembic · MongoDB · ChromaDB (vector DB) · Vercel · Render · Railway · Stripe · Strapi CMS · Supabase

**Testing & Quality** pytest (async) · CI-gated deploys · Offline eval · dependency-injected test isolation · Alembic migration review

**System Design (working knowledge):** Redis caching, load balancing, CDNs, message queues, read replicas, object storage, and AWS (ECS/RDS) — architectural understanding applied in design discussions and interview prep; not yet deployed in a production project.

## PROFESSIONAL EXPERIENCE

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### Full-Stack Developer | SwaPE IT Solutions *Dec 2023 – Present*

- Architected and shipped production web apps using Next.js (App Router), Node.js, React, and Python (FastAPI) services, with fully responsive UIs via Tailwind CSS and Framer Motion.
- Led system design decisions and defined scalable architecture patterns across multiple client projects.
- Integrated LLM APIs (OpenAI, Google Gemini) into client-facing features, gaining hands-on experience with prompt engineering and AI-powered product flows.
- Improved code quality through automated tooling, linting, peer reviews, and performance profiling.
- Integrated third-party services and APIs, reducing development time by establishing reusable integration patterns.

### Full-Stack Developer | Raj IT Solutions Ltd. *Nov 2021 – Dec 2023*

- Built RESTful APIs with Next.js, Node.js (Express), and Python (FastAPI); implemented real-time features using Socket.IO serving concurrent users without performance degradation.
- Integrated Strapi headless CMS, enabling non-technical stakeholders to manage content without developer intervention.
- Established CI/CD pipelines with Docker, cutting deployment time significantly and reducing production incidents.

### Freelance Full-Stack Developer | Independent *Oct 2015 – Present*

- Building and shipping AI-agent and AI-assisted products (see AI Projects below), spanning Next.js frontends, FastAPI/Node backends, and LangChain/LangGraph agent orchestration.
- Delivered e-commerce and booking systems using Node.js, React, and Next.js with Stripe payment integration, role-based access control, and RESTful APIs.
- Designed and optimised PostgreSQL, MongoDB, and MySQL databases for scalable, production-ready applications.

### Graphic Designer | Cloudville *Jan 2020 – Oct 2021*

- Designed digital assets and social media content; developed strong visual communication and client-facing skills directly applicable to frontend UI work.

## AI PROJECTS

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### Support Ops Copilot — Multi-Agent Customer Support System · *LangGraph · FastAPI · OpenAI API · ChromaDB · Next.js · MCP*

- Built a LangGraph multi-agent pipeline (triage → RAG → draft → guardrail → human approval → MCP execution) with genuine human-in-the-loop control via `interrupt()`, blocking any customer-facing action until human sign-off.
- Guardrails & security: independent LLM-as-judge guardrail + rule-based safety checks (prompt-injection detection, credential leakage) that auto-escalates any response failing verification instead of sending it.
- Offline eval harness gates CI on GitHub Actions, catching real regressions before they shipped.
- Integrated a real MCP client-server tool-calling architecture (not a mocked schema): a standalone MCP server exposed over stdio, with the action-executor calling it via the actual protocol handshake.
- Instrumented per-node token usage and latency across the pipeline, surfaced via the API for cost-per-resolution analysis.
- Applied cost-aware model routing (gpt-4o-mini for triage/guardrail, gpt-4o for drafting) and shipped via a documented small-team Git workflow (feature branches, PR reviews, CI-gated merges).

Live demo: [github.com/HillolPias/support-ops-copilot](https://github.com/HillolPias/support-ops-copilot)

### Ledger — AI-Powered Task & Project Manager · *FastAPI · Next.js · PostgreSQL · LangGraph · ChromaDB · LangSmith · Docker*

- Built a ReAct-style AI agent (LangGraph) with 17 tools for CRUD operations and RAG-based semantic search (ChromaDB), streaming responses token-by-token; traced end-to-end in LangSmith.
- Designed async REST API (FastAPI, SQLAlchemy 2.0) with Alembic migrations, deployed on PostgreSQL (Neon).
- Built a full CI/CD pipeline (GitHub Actions): an automated pytest suite against an ephemeral Postgres instance gates both Docker image builds (Docker Hub) and production deploys (Render).
- Diagnosed and resolved LLM tool-selection ambiguity and argument-mapping errors in the agent's tool-calling logic using LangSmith traces — failures invisible to standard error handling, since the agent returned confident but incorrect answers rather than throwing exceptions.
- Containerised with Docker and Docker Compose (healthchecks, named volumes, service networking) for reproducible local development.
- Built a custom-designed Next.js/TypeScript frontend with a live-streaming chat UI.

Live demo: [taskapi-fronend.vercel.app](https://taskapi-fronend.vercel.app) · Frontend: [github.com/HillolPias/taskapi-fronend](https://github.com/HillolPias/taskapi-fronend) · Backend: [github.com/HillolPias/taskapi-backend](https://github.com/HillolPias/taskapi-backend)

### Cookmind — AI Recipe Generator · *Next.js · TypeScript · Google Gemini API · Strapi · PostgreSQL · Clerk · Arcjet*

- Built an AI recipe generator that analyses a photo of a user's ingredients via the Google Generative AI (Gemini) API and returns tailored, ready-to-cook recipes, with drag-and-drop image upload and PDF export of results.
- Implemented Clerk-based authentication and Arcjet bot/rate-limit protection on a Next.js 16 (App Router) + React 19 frontend, backed by a Strapi 5 headless CMS on PostgreSQL.

Live demo: [cookmind-ivory.vercel.app](https://cookmind-ivory.vercel.app) · GitHub: [github.com/HillolPias/cookmind](https://github.com/HillolPias/cookmind)

## ADDITIONAL PROJECTS

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- Chatty App — real-time full-stack chat application (JavaScript). [GitHub](#)
- Product Store (MERN) — full-stack e-commerce product catalog. [GitHub](#)
- NextLevel Food/Foodies — recipe sharing application. [GitHub](#)

## EDUCATION

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BSc, Applied Physics & Electronic Engineering — University of Rajshahi

## COURSES & CERTIFICATIONS

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Agentic AI Engineering with LangChain & LangGraph (Udemy) · Functions, Tools & Agents with LangChain (DeepLearning.AI) · React — incl. Next.js, Redux (Udemy) · FastAPI Complete Course (Udemy) · Full-Stack Web Dev Bootcamp (Udemy) · CCNA — RUET